

**Math 2551 Worksheet: Multivariable Functions**

1. Find and sketch the domain for each function.

(a)  $f(x, y) = \sqrt{x - y - 1}$ .

(b)  $f(x, y) = \sqrt{(x - 4)(y^2 - 1)}$ .

(c)  $f(x, y) = \cos^{-1}(y - 4x^2)$ .

(d)  $f(x, y) = \frac{1}{4 - x^2 - y^2}$ .

(e)  $f(x, y) = \frac{1}{\ln(4 - x^2 - y^2)}$

2. Match the surfaces (a)-(g) with the written descriptions of the level curves (A)-(F). You may use each description once, multiple times, or not at all.

(a)  $z = 2x^2 + 3y^2$

(b)  $z = x^2 + y^2$

(c)  $z = \frac{1}{x - 1}$

(d)  $z = 2x + 3y$

(e)  $z = \sqrt{25 - x^2 - y^2}$

(f)  $z = \sqrt{x^2 + y^2}$

(g)  $z = xy$ .

(B) a collection of unequally spaced concentric circles

(B) a collection of unequally spaced parallel lines

(B) a collection of equally spaced concentric circles

(B) two straight lines and a collection of hyperbolas

(B) a collection of concentric ellipses

(B) a collection of equally spaced parallel lines

3. Find an equation for the level curve of the function  $F(x, y) = \frac{2y - x}{x + y + 1}$  passing through  $(-1, 1)$ .
4. Find the equation for the level surface of the function  $f(x, y, z) = \sqrt{x^2 + y^2 + z^2}$  passing through  $(1, 1, 1)$ .
5. Let  $f(x, y) = (x - y)^2$ . Determine the equations and shapes of the cross-sections when  $x = 0$ ,  $y = 0$ , and  $x = y$ , and describe the level curves. Use this information to produce a sketch of the graph of the surface. Confirm your sketch using a 3d graphing utility.